

SCHOOL OF SCIENCE & TECHNOLOGY

COURSE: DSA 4020A Natural Language Processing

Document Title: Semester Project

Semester Project: Machine Translation of Public Service Announcements (PSAs)

Project Description: Develop a proof-of-concept multilingual machine translation (MT) system for Public Service Announcements (PSAs) in Kenya. The system translates between English/Kiswahili and selected under-resourced indigenous languages (e.g., Ekegusii – Bantu, Dholuo – Nilotic, Somali – Cushitic), making government information more accessible. The final deliverable is a deployable digital public good (web/mobile app) demonstrating few-shot cross-lingual transfer learning on a curated PSA dataset.

PSAs: PSAs are short, clear, action-oriented, and sometimes urgent information to inform, warn, or guide the public about an action they should take (e.g., health measures, safety advisories, deadlines, disaster alerts) and the tone is often directive or advisory. E.g. *“IEBC reminds voters to verify their details via SMS.”* *“Ministry of Health: Avoid unnecessary travels to Ebola hotspots”*

Project Duration: 4 weeks

Team: 4–5 students

Tools: Python (Hugging Face, pandas, BeautifulSoup), evaluation libraries, Streamlit/Gradio for deployment.

Sub-Objectives

1. Curate a high-quality multilingual dataset ($\geq 5,000$ sentences per language pair).
2. Explore few-shot cross-lingual transfer using pre-trained models (mT5, NLLB, mBART, etc.).
3. Evaluate the model for accuracy and cultural appropriateness.
4. Deploy the model as a digital public good.

Week 1: Data Collection & Curation (Focus: Sub-objective 1)

Main Deliverables: Parallel dataset of $\geq 5,000$ sentences.

Milestone Checklist

- ☐ Identify and document ≥ 10 reliable sources (gov websites, X/Twitter official accounts, media archives, NGOs).
- ☐ Implement hybrid scraping pipeline (manual + automated with BeautifulSoup/Selenium; respect robots.txt and rate limits).

- ❑ Collect raw PSAs across key domains (Education, Health, Security, Agriculture, Governance).
- ❑ Create structured dataset (CSV/JSON) with columns: PSA_ID, Domain, English, Kiswahili, Target Languages (placeholders), Source, Date, Metadata.
- ❑ Perform initial cleaning: deduplication, language detection (langdetect/fasttext), relevance filtering.
- ❑ Reach $\geq 5,000$ parallel sentences (or equivalent per pair) with basic validation.
- ❑ Submit Week 1 report: dataset summary stats + sample entries + challenges faced.
- ❑ Team meeting: Assign roles and review ethical scraping practices.

Success Criteria: Dataset uploaded to GitHub; supervisor approves quality/diversity.

Week 2: Data Processing & Exploratory Data Analysis (EDA) (Sub-objectives 1 & 2)

Main Deliverables: Cleaned dataset + EDA report.

Milestone Checklist:

- ❑ Build preprocessing pipeline (tokenization, normalization, code-switching handling, glossary for cultural terms).
- ❑ Conduct full EDA: domain distribution (pie/bar charts), text length histograms, vocabulary size, language-pair statistics.
- ❑ Create native-speaker validation subset (~500 sentences) and collect feedback.
- ❑ Handle issues: missing translations, orthographic variations, domain imbalance.
- ❑ Version-control the final cleaned dataset (use DVC or Git LFS if large).
- ❑ Produce EDA notebook/report with visualizations and insights.
- ❑ Prepare data splits (train/dev/test) for modeling.
- ❑ Week 2 team review + supervisor check-in on data quality.

Success Criteria: EDA visuals and cleaned dataset ready; preprocessing code reusable.

Week 3: Modeling with Transfer Learning (Focus: Sub-objective 2)

Main Deliverables: Trained baseline + fine-tuned models.

Milestone Checklist:

- ❑ Set up experiment tracking (Weights & Biases or MLflow).
- ❑ Implement ≥ 2 pre-trained models (e.g., mT5-small, NLLB-200 distilled, mBART) with few-shot fine-tuning.
- ❑ Train models on curated dataset (handle low-resource setting with techniques like freezing layers or data augmentation).
- ❑ Run ablation studies (e.g., zero-shot vs few-shot, domain adaptation).
- ❑ Save model checkpoints and logs.
- ❑ Document hyperparameters, training time, and preliminary results.
- ❑ Mid-week check-in: Troubleshoot GPU/Colab issues if any.

- Week 3 deliverable: Model inference script + initial performance summary.

Success Criteria: Working translation demo (command-line or notebook) for sample PSAs.

Week 4: Evaluation, Deployment & Documentation (Focus: Sub-objectives 3 & 4)

Main Deliverables: Evaluated model + live demo + final report.

Milestone Checklist:

- Compute automatic metrics (BLEU, chrF, COMET, SacreBLEU).
- Conduct human evaluation (fluency, adequacy, cultural accuracy) with native speakers (use 100+ test sentences).
- Perform error analysis and document limitations.
- Deploy model as web app (Streamlit/Gradio) or mobile prototype: input PSA → select target language → output translation.
- Add features: example PSAs, confidence scores, feedback form.
- Prepare final GitHub repo: code, dataset (or link), notebooks, README, license (e.g., CC-BY).
- Write final report/presentation covering all sub-objectives, results, and impact.
- Demo day: Present live translation of real PSAs + Q&A.

Success Criteria: Publicly accessible demo; report submitted; all code reproducible.

Overall Rubric (100 points)

- Sub-objective 1 (Dataset): 25 pts
- Sub-objective 2 (Modeling): 30 pts
- Sub-objective 3 (Evaluation): 20 pts
- Sub-objective 4 (Deployment): 15 pts
- Overall Quality, Documentation & Presentation: 10 pts